UNIVERSITY OF ASIA PACIFIC

Department of Computer Science and Engineering

Project

Shopping Cart System

Course Title – Object oriented programming 2

Course Code – 202

Submitted by:

Registration no. – 23201143, 23201156, 23201157

Name – Muhammad **Mehedi** Hasan(143), Al Shahriar Hossain **Piyal**(156), Md. **Alvi** Sarwar (157)

Section – C2

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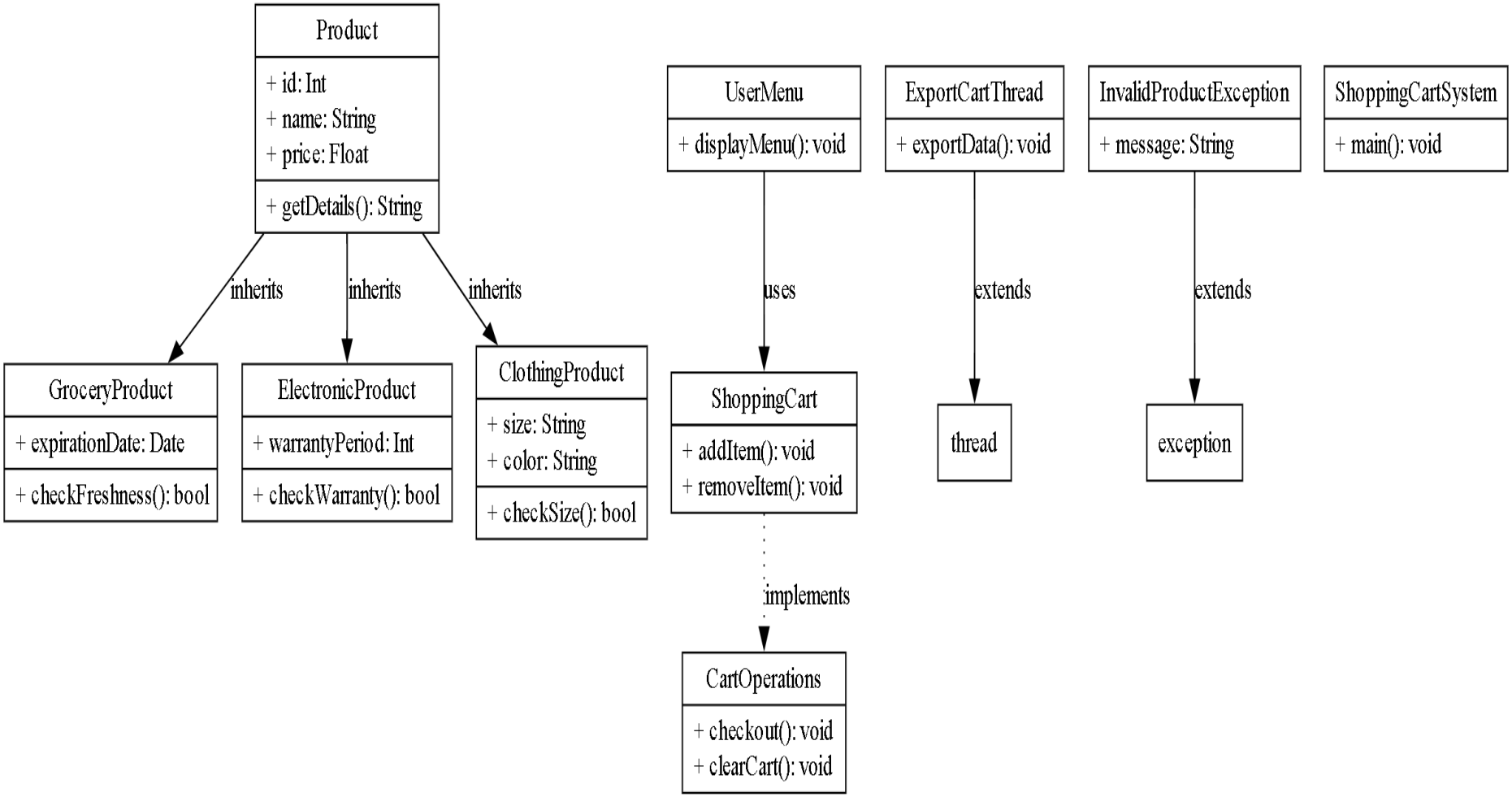
Submitted to:

Baivab Das

Lecturer,CSE,UAP

The Class Diagram



```java
// Product.java
import java.io.Serializable;

// Abstract base class for all product types
public abstract class Product implements Serializable {
    protected String id;
    protected String name;
    protected double price;

    // Constructor to initialize basic product properties
    public Product(String id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    // Abstract method to get product details (to be overridden)
    public abstract String getDetails();

    public String getId() { return id; }
    public double getPrice() { return price; }
}

// GroceryProduct.java
// Represents a grocery item with weight
public class GroceryProduct extends Product {
    private double weight;

    public GroceryProduct(String id, String name, double price, double weight) {
```

```
        super(id, name, price);
        this.weight = weight;
    }

    @Override
    public String getDetails() {
        return "[Grocery] " + id + ", " + name + ", Rs." + price + ", " + weight + "kg";
    }
}
```

// ElectronicProduct.java

// Represents an electronic item with warranty information

```
public class ElectronicProduct extends Product {
    private int warrantyMonths;

    public ElectronicProduct(String id, String name, double price, int warrantyMonths) {
        super(id, name, price);
        this.warrantyMonths = warrantyMonths;
    }

    @Override
    public String getDetails() {
        return "[Electronic] " + id + ", " + name + ", Rs." + price + ", Warranty: " + warrantyMonths + "
months";
    }
}
```

// ClothingProduct.java

// Represents a clothing item with size

```
public class ClothingProduct extends Product {
    private String size;

    public ClothingProduct(String id, String name, double price, String size) {
        super(id, name, price);
        this.size = size;
    }

    @Override
    public String getDetails() {
        return "[Clothing] " + id + ", " + name + ", Rs." + price + ", Size: " + size;
    }
}

// CartOperations.java
// Interface that defines basic cart operations
public interface CartOperations {
    void addItem(Product product);
    void removeItem(String productId);
    void displayCart();
    double calculateTotal();
}

// ShoppingCart.java
import java.util.*;

// Implementation of CartOperations using a List
public class ShoppingCart implements CartOperations {
    private List<Product> cart;
```

```
public ShoppingCart() {
    cart = new ArrayList<>();
}

// Add product to cart
public void addItem(Product product) {
    cart.add(product);
    System.out.println(product.getId() + " added to cart.");
}

// Remove product from cart by ID
public void removeItem(String productId) {
    cart.removeIf(p -> p.getId().equals(productId));
    System.out.println(productId + " removed from cart.");
}

// Display all items in the cart
public void displayCart() {
    System.out.println("\nItems in Cart:");
    for (Product p : cart) {
        System.out.println(p.getDetails());
    }
}

// Calculate total price of items in cart
public double calculateTotal() {
    return cart.stream().mapToDouble(Product::getPrice).sum();
}
```

```
    // Return the list of cart items (for exporting)
    public List<Product> getCartItems() {
        return cart;
    }
}

// ExportCartThread.java
import java.io.*;

// Thread to export cart data to a text file
public class ExportCartThread extends Thread {
    private ShoppingCart cart;
    private String filename;

    public ExportCartThread(ShoppingCart cart, String filename) {
        this.cart = cart;
        this.filename = filename;
    }

    @Override
    public void run() {
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(filename))) {
            for (Product product : cart.getCartItems()) {
                writer.write(product.getDetails());
                writer.newLine();
            }
            writer.write("Total: Rs." + cart.calculateTotal());
            System.out.println("\nCart exported to " + filename);
```

```
        } catch (IOException e) {
            System.err.println("File writing error: " + e.getMessage());
        }
    }
}

// InvalidProductException.java
// Custom exception for invalid product operations
public class InvalidProductException extends Exception {
    public InvalidProductException(String message) {
        super(message);
    }
}

// UserMenu.java
import java.util.Scanner;

// Handles user interaction and menu operations
public class UserMenu {
    private ShoppingCart cart;
    private Scanner scanner;

    public UserMenu() {
        cart = new ShoppingCart();
        scanner = new Scanner(System.in);
    }

    // Display menu and handle user input
    public void start() {
```

```
    boolean exit = false;

    while (!exit) {
        System.out.println("\n--- Shopping Cart Menu ---");
        System.out.println("1. Add Grocery Product");
        System.out.println("2. Add Electronic Product");
        System.out.println("3. Add Clothing Product");
        System.out.println("4. Remove Product by ID");
        System.out.println("5. View Cart");
        System.out.println("6. Export Cart to File");
        System.out.println("7. Exit");
        System.out.print("Choose an option: ");

        int choice = scanner.nextInt();
        scanner.nextLine();

        try {
            switch (choice) {
                case 1:
                    System.out.print("Enter ID, Name, Price, Weight (kg): ");
                    Product g = new GroceryProduct(scanner.next(), scanner.next(), scanner.nextDouble(),
scanner.nextDouble());
                    cart.addItem(g);
                    break;
                case 2:
                    System.out.print("Enter ID, Name, Price, Warranty (months): ");
                    Product e = new ElectronicProduct(scanner.next(), scanner.next(), scanner.nextDouble(),
scanner.nextInt());
                    cart.addItem(e);
```

```
                break;
            case 3:
                System.out.print("Enter ID, Name, Price, Size: ");
                Product c = new ClothingProduct(scanner.next(), scanner.next(), scanner.nextDouble(),
scanner.next());
                cart.addItem(c);
                break;
            case 4:
                System.out.print("Enter Product ID to remove: ");
                cart.removeItem(scanner.next());
                break;
            case 5:
                cart.displayCart();
                System.out.println("Total: Rs." + cart.calculateTotal());
                break;
            case 6:
                ExportCartThread thread = new ExportCartThread(cart, "cart_summary.txt");
                thread.start();
                break;
            case 7:
                exit = true;
                break;
            default:
                System.out.println("Invalid option. Try again.");
        }
    } catch (Exception ex) {
        System.err.println("Error: " + ex.getMessage());
    }
}
```

```
        scanner.close();
        System.out.println("Exiting Shopping Cart System.");
    }
}
```

// ShoppingCartSystem.java

// Entry point of the shopping cart application

```
public class ShoppingCartSystem {
    public static void main(String[] args) {
        UserMenu menu = new UserMenu();
        menu.start();
    }
}
```

END OF CODE

I/O ScreenShots

```
public static void main(String[] a
Console ×
ShoppingCartSystem (1) [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.2.v2

--- Shopping Cart Menu ---
1. Add Grocery Product
2. Add Electronic Product
3. Add Clothing Product
4. Remove Product by ID
5. View Cart
6. Export Cart to File
7. Exit
Choose an option: 1
Enter ID, Name, Price, Weight (kg): 123
alu
20
1.5
123 added to cart.
```

```
--- Shopping Cart Menu ---
1. Add Grocery Product
2. Add Electronic Product
3. Add Clothing Product
4. Remove Product by ID
5. View Cart
6. Export Cart to File
7. Exit
Choose an option: 2
Enter ID, Name, Price, Warranty (months): 131
telephone
1200
12
131 added to cart.
```

```
--- Shopping Cart Menu ---
1. Add Grocery Product
2. Add Electronic Product
3. Add Clothing Product
4. Remove Product by ID
5. View Cart
6. Export Cart to File
7. Exit
Choose an option: 3
Enter ID, Name, Price, Size: 151
undwerwear
120
xl
151 added to cart.
```

```
Console
ShoppingCartSystem (1) [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.2.v20250131-0604\jre\bin\javaw.exe (Jun 2, 202

--- Shopping Cart Menu ---
1. Add Grocery Product
2. Add Electronic Product
3. Add Clothing Product
4. Remove Product by ID
5. View Cart
6. Export Cart to File
7. Exit
Choose an option: 5

Items in Cart:
[Grocery] 123, alu, Rs.20.0, 1.5kg
[Electronic] 131, telephone, Rs.1200.0, Warranty: 12 months
[Clothing] 151, undwerwear, Rs.120.0, Size: xl
Total: Rs.1340.0
```

```
Choose an option: 6

--- Shopping Cart Menu ---
1. Add Grocery Product
2. Add Electronic Product
3. Add Clothing Product
4. Remove Product by ID
5. View Cart
6. Export Cart to File
7. Exit
Choose an option:
Cart exported to cart_summary.txt
7
Exiting Shopping Cart System.
```